

Write legibly showing all the steps VERY clearly. (Total points = 100).

- (1) (a) Define the following collections: • semi-algebra, • algebra, • σ -algebra.
 (b) A nonempty collection $\mathcal{M}$ is said to be a *monotone class* if it is closed under increasing unions and decreasing intersections (i.e. If $\{A_n\}$ is an increasing sequence of sets in $\mathcal{M}$ then $\cup_{n \geq 1} A_n \in \mathcal{M}$ and if $\{A_n\}$ is a decreasing sequence of sets in $\mathcal{M}$ then $\cap_{n \geq 1} A_n \in \mathcal{M}$). Prove that if $\mathcal{F}$ is a σ -algebra, it is also a monotone class.
 (c) Let $\Omega = \{a, b, c\}$ and $\mathcal{F} = \{\emptyset, \{a\}, \{b\}, \{c\}, \Omega\}$ be a collection. Is it a monotone class? Is it an σ -algebra? Justify your answer.

(5×3+5+(5+5) = 30 points)

- (2) (a) Let Ω be an uncountable set and let $\mathcal{F}$ be the countable-cocountable collection, i.e. $\mathcal{F} = \{A \subset \Omega : A \text{ countable or } A^c \text{ countable}\}$. Define a real-valued function μ on $\mathcal{F}$ as follows: $\mu(A) = 1$ if A is uncountable and $\mu(A) = 0$ if A countable. Show that μ is a *probability* measure on $(\Omega, \mathcal{F})$.
 (b) Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mu : \mathcal{F} \rightarrow [0, 1]$ be a set function satisfying the conditions : (i) μ is *finitely* additive on $\mathcal{F}$, (ii) $\mu(\Omega) = 1$, and (iii) if $E_n \in \mathcal{F}, n \geq 1$ are disjoint with $\cup_{n=1}^{\infty} E_n = \Omega$, then $\sum_{n=1}^{\infty} \mu(E_n) = 1$.
 Show that μ is a *probability* measure on $(\Omega, \mathcal{F})$.

(2×10 = 20 points)

- (3) Take $F(x) = \frac{1}{3}\mathbf{I}_{(0,x]}(0) + \frac{2}{3}\mathbf{I}_{(0,x]}(1), x \in \mathbb{R}$.
 (a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.
 [Write down the steps of the theorems carefully and state the results used in each step]
 (b) Why is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique? What is the name of this measure?
 (c) If $A = \{x \in \mathbb{R} : |x| > \frac{1}{2}\}$, then compute $\mu_F^*(A)$.

(3 × 10 = 30 points)

- (4) (a) Give an example of two (distinct) functions f and g that are **not** $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable, but $f + g$ is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
 (b) Let $\Omega_1 = \mathbb{R}$ and $\mathcal{F}_1 = \{\emptyset, A, A^c, \Omega_1\}$, with $A = (0, 5]$. Let $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $T : \Omega_1 \rightarrow \Omega_2$ be defined as

$$T(x) = \begin{cases} 5, & \text{if } x \leq 0 \\ 0, & \text{if } x > 0 \end{cases}$$

Is T $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable? Justify your answer.

(2×10 = 20 points)